CSA5801 DevSecOps Engineering and Application Security

Name:Vickramabharathi V
Reg. No.:192465069

Experiment 1

Installation and Configuration of Git, GitHub, RStudio and Required DevSecOps Tools

R Code:

```
grViz("
digraph G {
  rankdir=LR
  Git -> GitHub -> RStudio -> DevSecOps_Tools
}
")
```

Output:

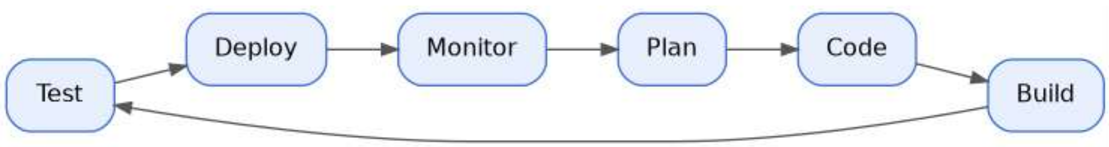
Experiment 2

Visualization of the DevSecOps Lifecycle using Workflow Modelling

R Code:

```
grViz("
digraph G {
  rankdir=LR
  Plan -> Code -> Build -> Test -> Deploy -> Monitor -> Plan
}
")
```

Output:



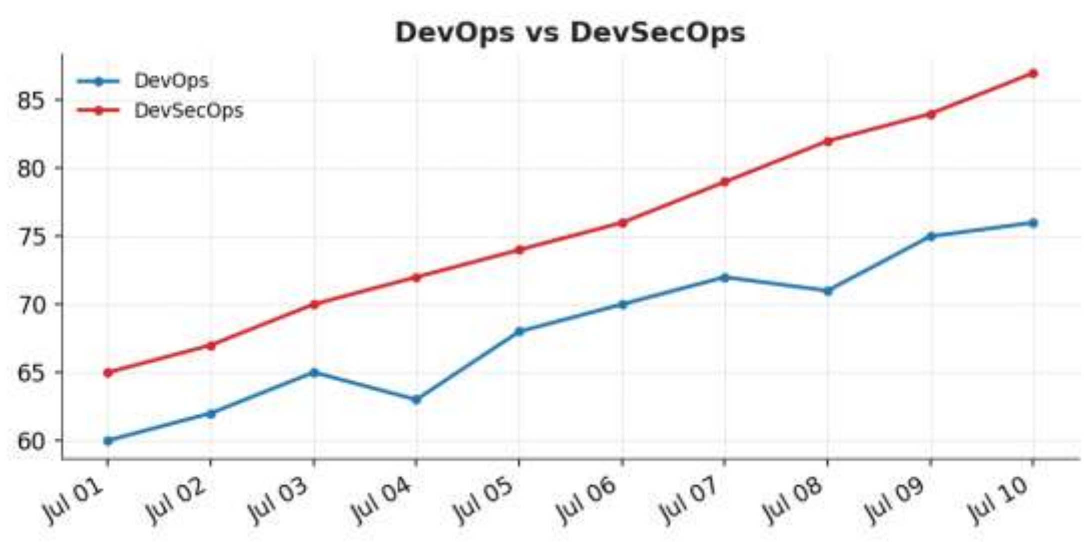
Experiment 3

Comparative Analysis of DevOps and DevSecOps using Synthetic Data

R Code:

```
d <- 1:10
x <- xts(cbind(
  DevOps = c(60,62,65,63,68,70,72,71,75,76),
  DevSecOps = c(65,67,70,72,74,76,79,82,84,87)
), order.by=as.Date("2026-07-01")+d-1)
dygraph(x, main="DevOps vs DevSecOps")
```

Output:



Experiment 4

Security Risk Assessment and Risk Matrix Simulation

R Code:

```
x <- xts(cbind(
  Risk = c(20,35,45,30,55,40,25,60,50,30)
), order.by=as.Date("2026-07-01")+0:9)
dygraph(x, main="Security Risk Simulation")
```

Output:

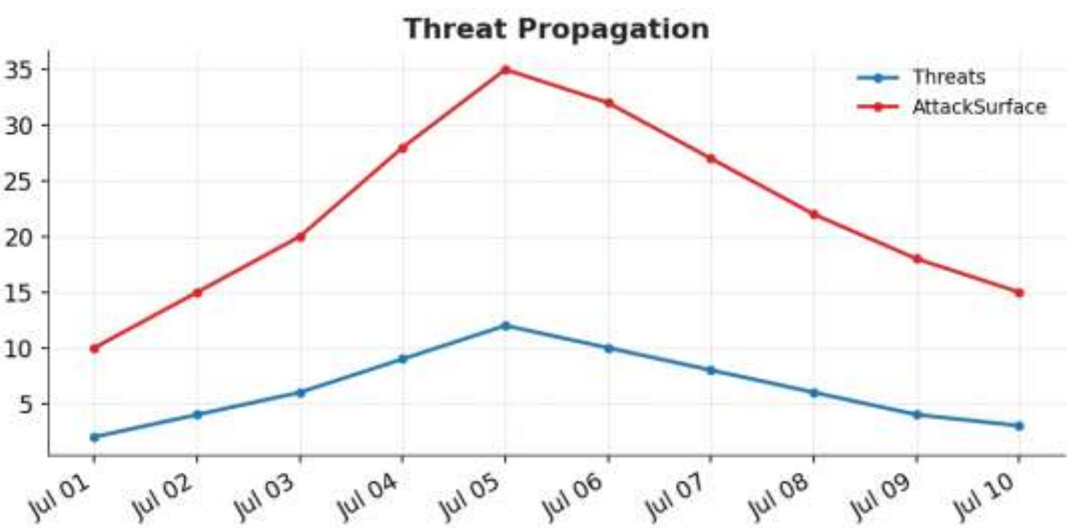
Experiment 5

Threat Propagation Simulation and Attack Surface Analysis

R Code:

```
x <- xts(cbind(
  Threats = c(2,4,6,9,12,10,8,6,4,3),
  AttackSurface = c(10,15,20,28,35,32,27,22,18,15)
), order.by=as.Date("2026-07-01")+0:9)
dygraph(x, main="Threat Propagation")
```

Output:



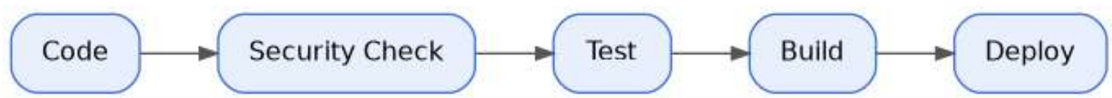
Experiment 6

Design and Simulation of Shift-Left Security Workflow

R Code:

```
grViz("
digraph G {
  rankdir=LR
  Code -> Security_Check -> Test -> Build -> Deploy
}
")
```

Output:



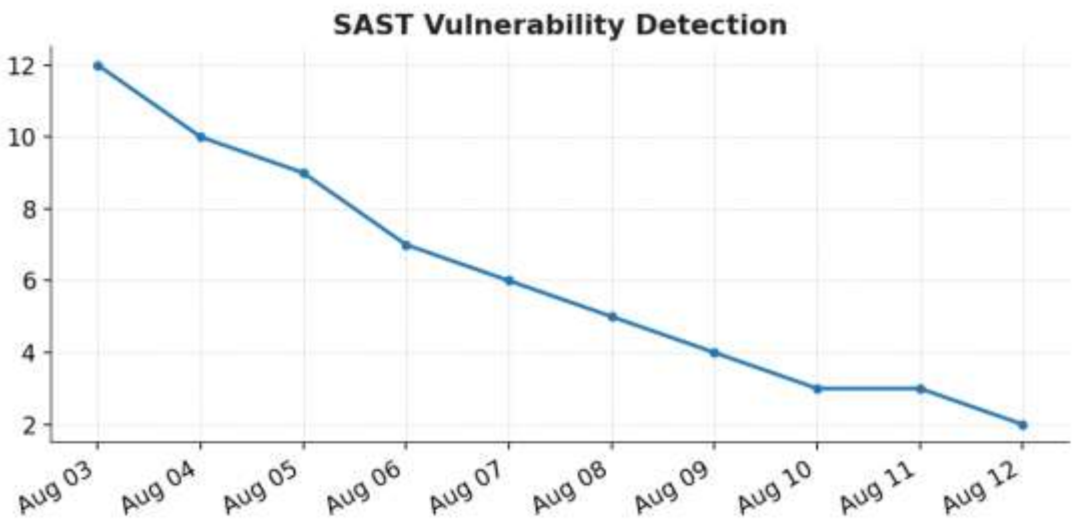
Experiment 7

Static Application Security Testing (SAST) and Security Verification

R Code:

```
x <- xts(cbind(
  Vulnerabilities = c(12,10,9,7,6,5,4,3,3,2)
), order.by=as.Date("2026-08-03")+0:9)
dygraph(x, main="SAST Vulnerability Detection")
```

Output:



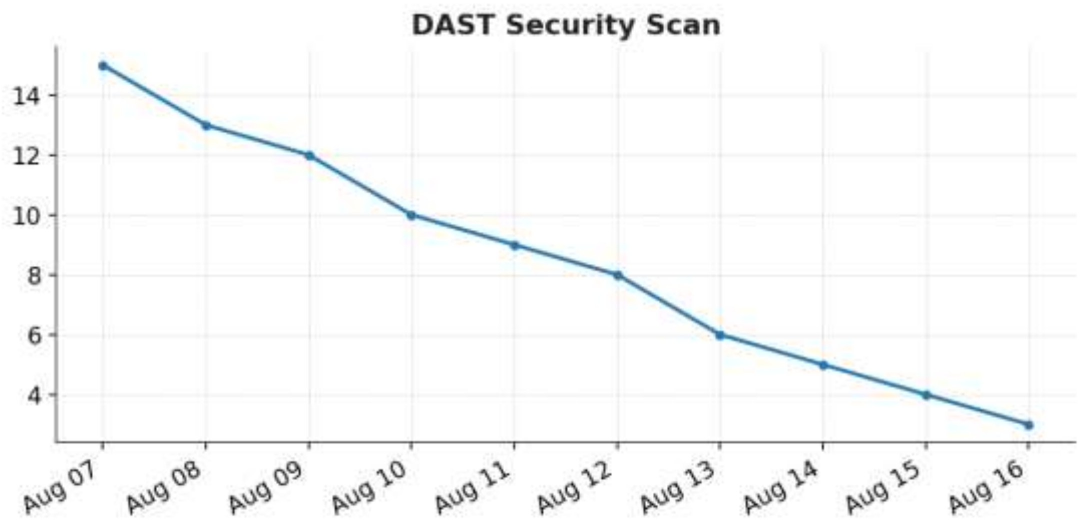
Experiment 8

Dynamic Application Security Testing (DAST)

R Code:

```
x <- xts(cbind(
  Vulnerabilities = c(15,13,12,10,9,8,6,5,4,3)
), order.by=as.Date("2026-08-07")+0:9)
dygraph(x, main="DAST Security Scan")
```

Output:



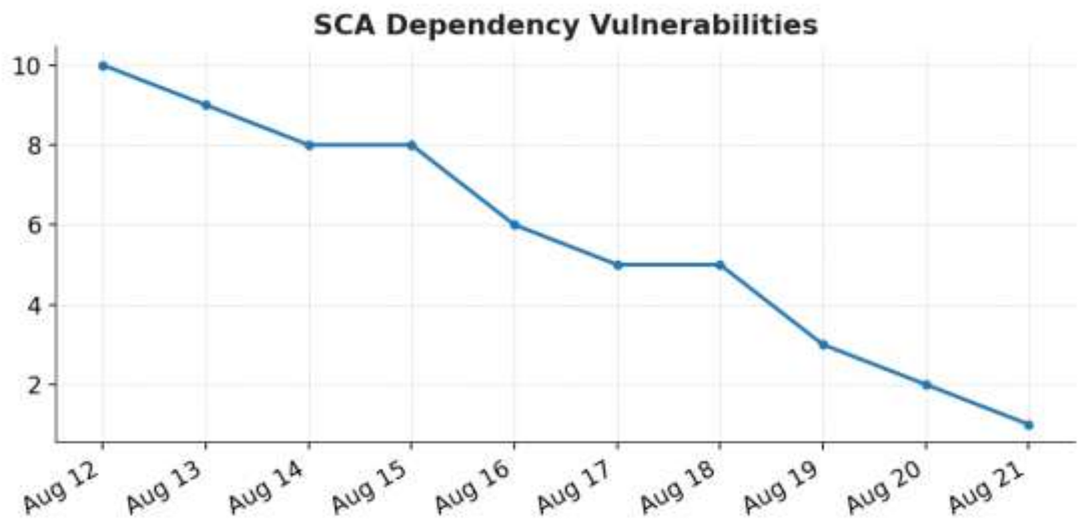
Experiment 9

Software Composition Analysis (SCA)

R Code:

```
x <- xts(cbind(
  VulnerableDependencies = c(10,9,8,8,6,5,5,3,2,1)
), order.by=as.Date("2026-08-12")+0:9)
dygraph(x, main="SCA Dependency Vulnerabilities")
```

Output:



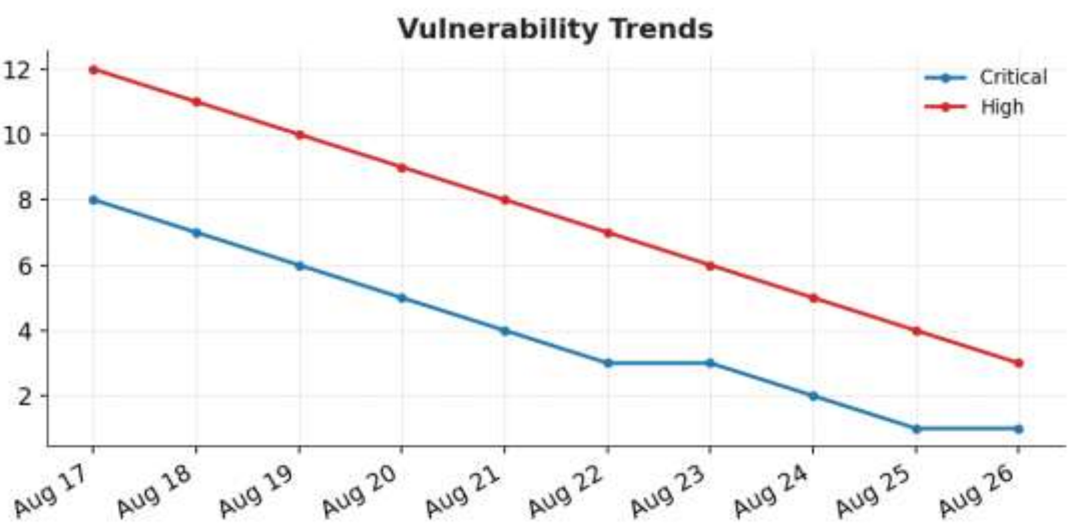
Experiment 10

Vulnerability Trend Analysis and Security Reporting

R Code:

```
x <- xts(cbind(
  Critical = c(8,7,6,5,4,3,3,2,1,1),
  High = c(12,11,10,9,8,7,6,5,4,3)
), order.by=as.Date("2026-08-17")+0:9)
dygraph(x, main="Vulnerability Trends")
```

Output:



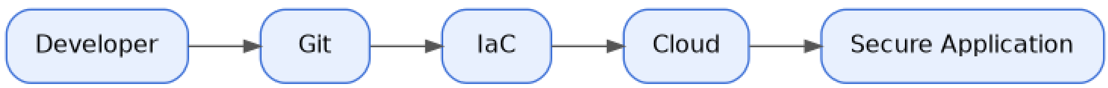
Experiment 11

Secure Infrastructure Architecture Design using IaC Concepts

R Code:

```
grViz("
digraph G {
  rankdir=LR
  Developer -> Git -> IaC -> Cloud -> Secure_Application
}
")
```

Output:



Experiment 12

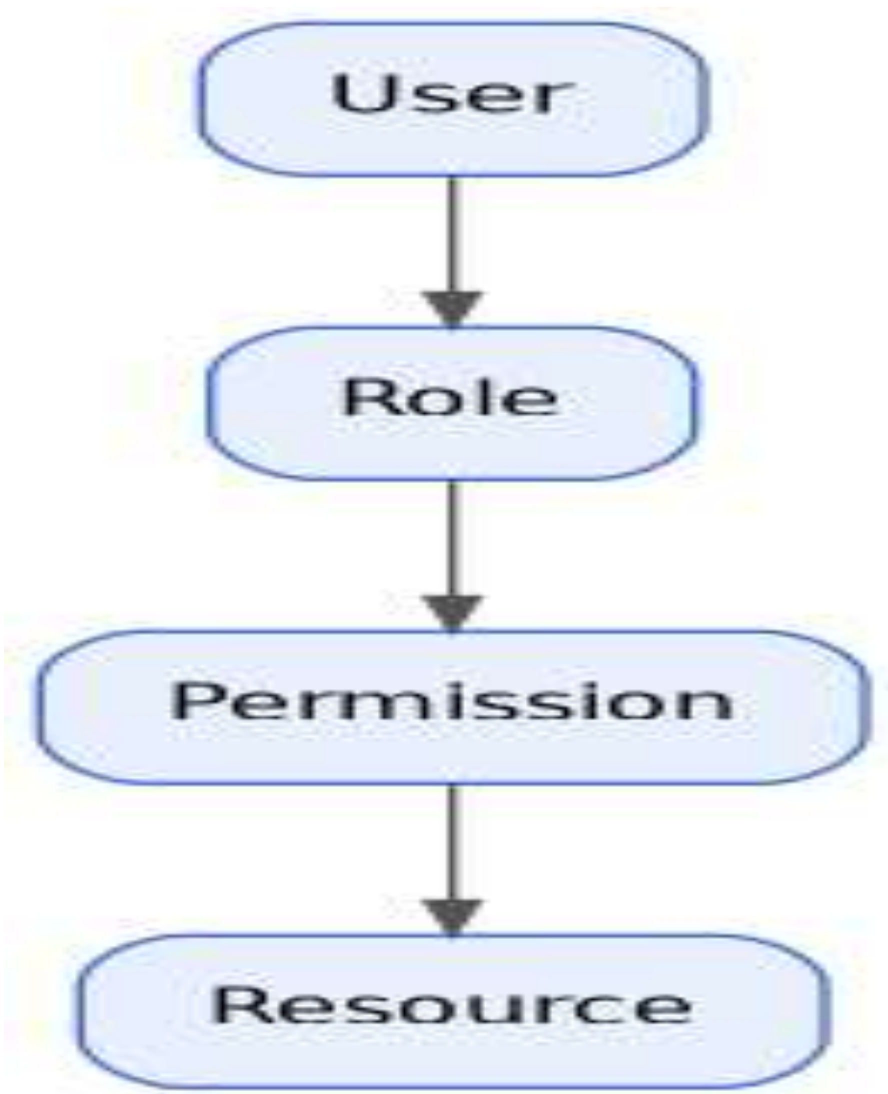
Identity and Access Management (IAM) Relationship Modelling

Type: Static → DiagrammeR

R Code:

```
grViz("
digraph G {
  User -> Role -> Permission -> Resource
}
")
```

Output:



Experiment 13

GitOps Workflow Design and Secure Deployment Simulation

R Code:

```
grViz("
digraph G {
  rankdir=LR
  Developer -> Git -> CI -> Security_Check -> CD -> Kubernetes
}
")
```

Output:

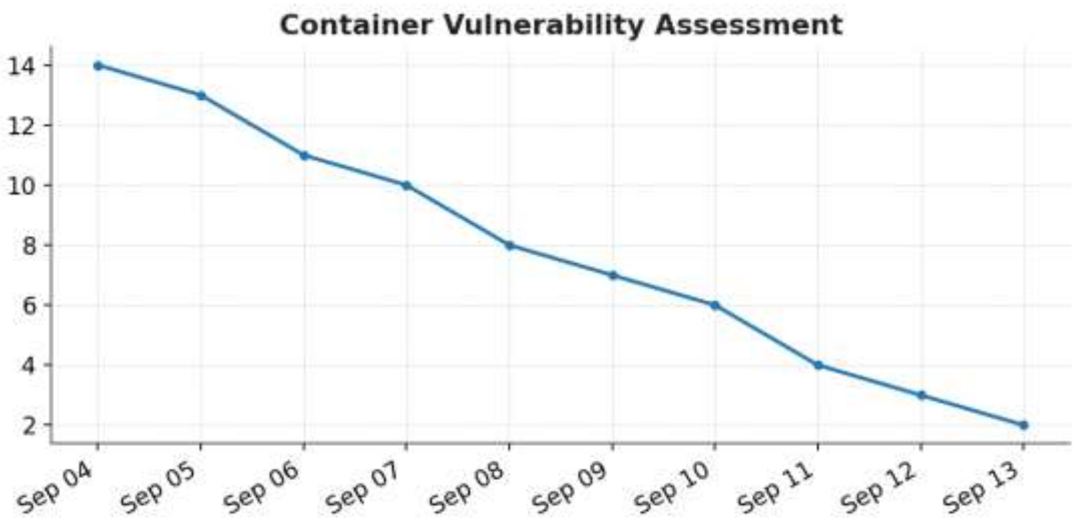
Experiment 14

Container Security and Vulnerability Assessment

R Code:

```
x <- xts(cbind(
  Vulnerabilities = c(14,13,11,10,8,7,6,4,3,2)
), order.by=as.Date("2026-09-04")+0:9)
dygraph(x, main="Container Vulnerability Assessment")
```

Output:



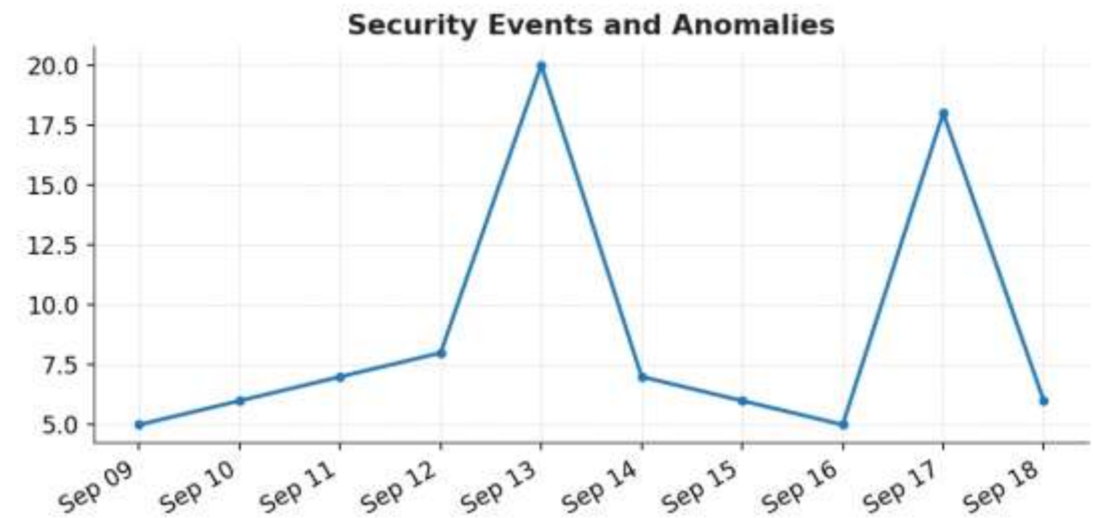
Experiment 15

Security Log Generation, Incident Timeline Analysis and Anomaly Detection

R Code:

```
x <- xts(cbind(
  Events = c(5,6,7,8,20,7,6,5,18,6)
), order.by=as.Date("2026-09-09")+0:9)
dygraph(x, main="Security Events and Anomalies")
```

Output:



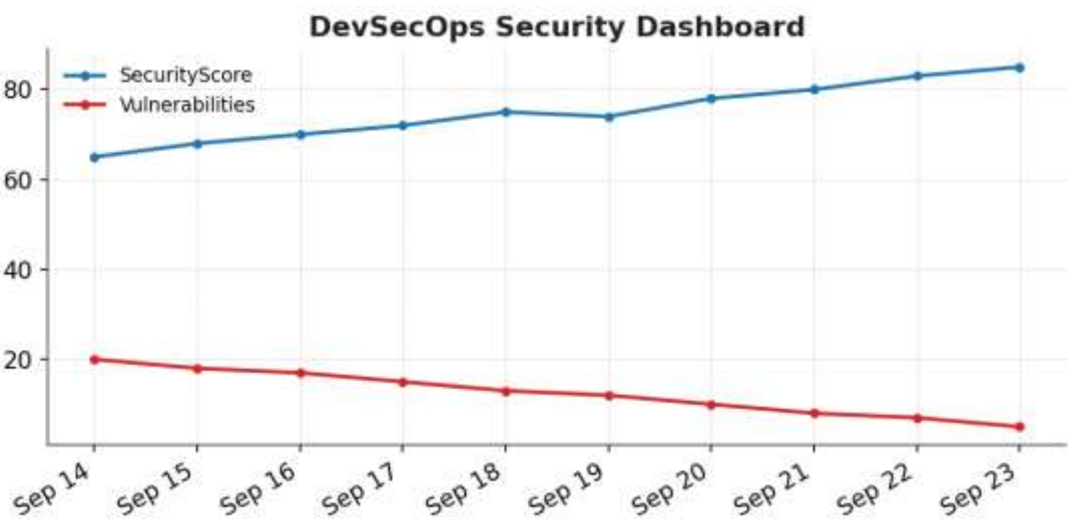
Experiment 16

Development of a DevSecOps Security Monitoring Dashboard

R Code:

```
x <- xts(cbind(
  SecurityScore = c(65,68,70,72,75,74,78,80,83,85),
  Vulnerabilities = c(20,18,17,15,13,12,10,8,7,5)
), order.by=as.Date("2026-09-14")+0:9)
dygraph(x, main="DevSecOps Security Dashboard")
```

Output:



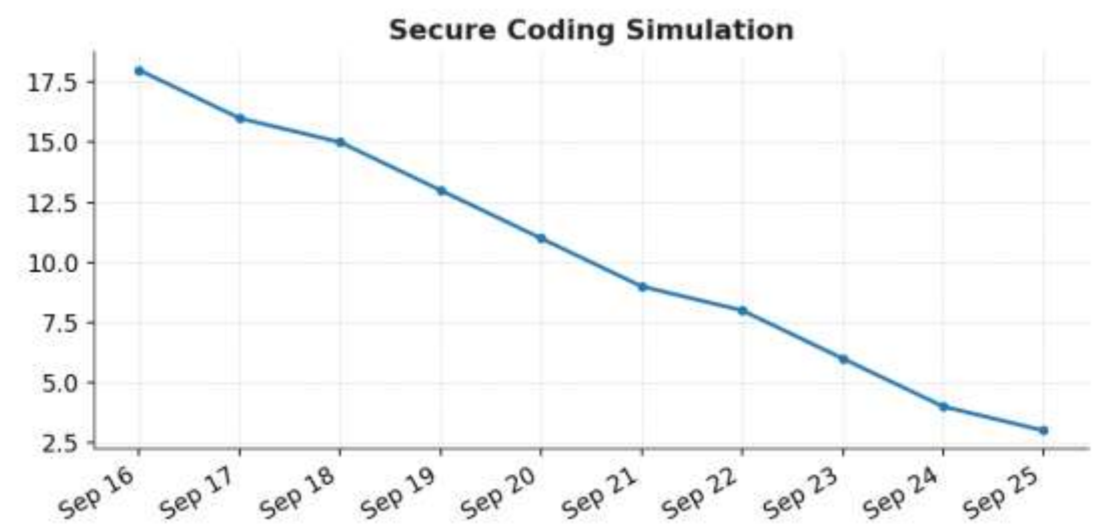
Experiment 17

Secure Coding Practices and Vulnerability Detection Simulation

R Code:

```
x <- xts(cbind(
  Vulnerabilities = c(18,16,15,13,11,9,8,6,4,3)
), order.by=as.Date("2026-09-16")+0:9)
dygraph(x, main="Secure Coding Simulation")
```

Output:



Experiment 18

OWASP Top 10 Vulnerability Identification and Risk Classification

R Code:

```
grViz("
digraph G {
  rankdir=LR
  Application -> OWASP_Top_10 -> Risk_Classification
}
")
```

Output:

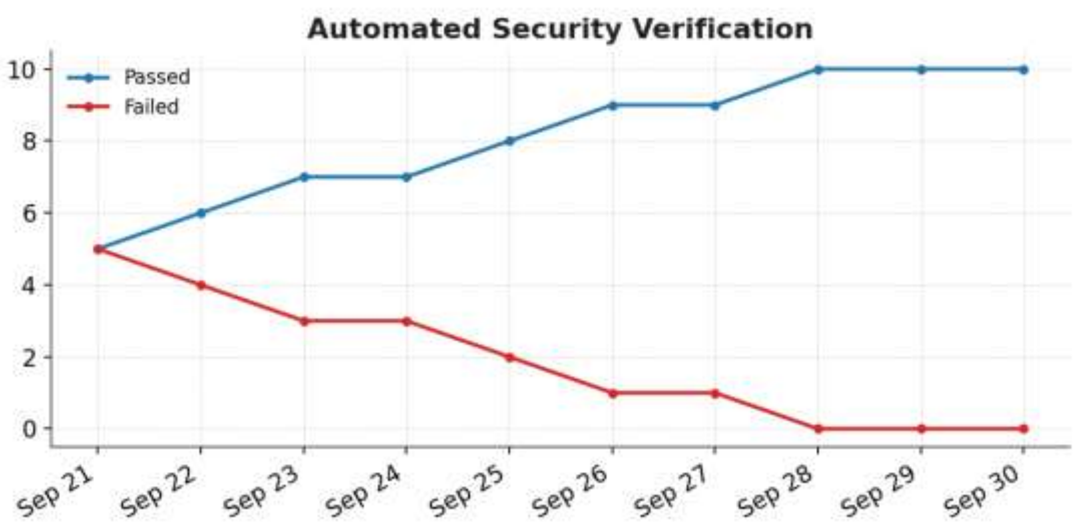
Experiment 19

Security Test Case Design and Automated Verification Simulation

R Code:

```
x <- xts(cbind(
  Passed = c(5,6,7,7,8,9,9,10,10,10),
  Failed = c(5,4,3,3,2,1,1,0,0,0)
), order.by=as.Date("2026-09-21")+0:9)
dygraph(x, main="Automated Security Verification")
```

Output:



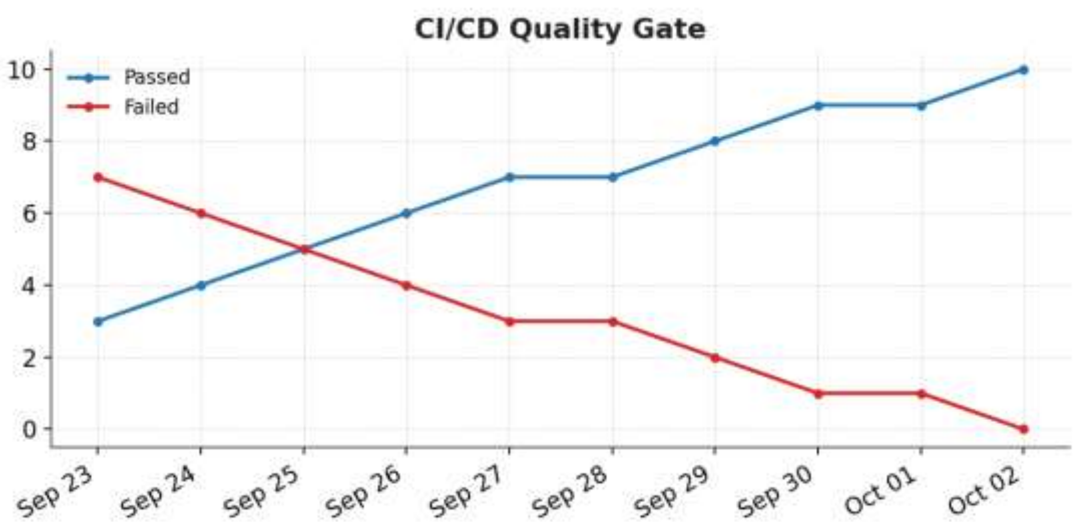
Experiment 20

CI/CD Pipeline Security Verification and Quality Gate Simulation

R Code:

```
x <- xts(cbind(
  Passed = c(3,4,5,6,7,7,8,9,9,10),
  Failed = c(7,6,5,4,3,3,2,1,1,0)
), order.by=as.Date("2026-09-23")+0:9)
dygraph(x, main="CI/CD Quality Gate")
```

Output:



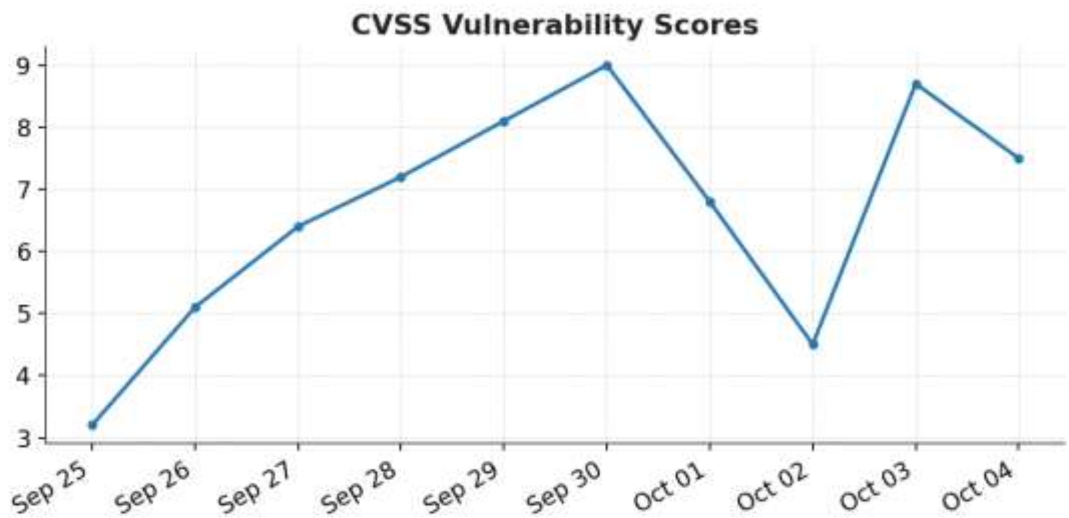
Experiment 21

Security Vulnerability Prioritization using CVSS-Based Scoring

R Code:

```
x <- xts(cbind(
  CVSS = c(3.2,5.1,6.4,7.2,8.1,9.0,6.8,4.5,8.7,7.5)
), order.by=as.Date("2026-09-25")+0:9)
dygraph(x, main="CVSS Vulnerability Scores")
```

Output:



Experiment 22

Secrets Detection and Secure Secrets Management Simulation

R Code:

```
x <- xts(cbind(
  SecretsFound = c(12,10,9,8,6,5,4,3,2,1)
), order.by=as.Date("2026-09-28")+0:9)
dygraph(x, main="Secrets Detection")
```

Output:

Experiment 23

API Security Testing and Vulnerability Analysis

R Code:

```
x <- xts(cbind(
  Vulnerabilities = c(10,9,8,7,6,5,4,3,2,1)
), order.by=as.Date("2026-09-30")+0:9)
dygraph(x, main="API Security Testing")
```

Output:

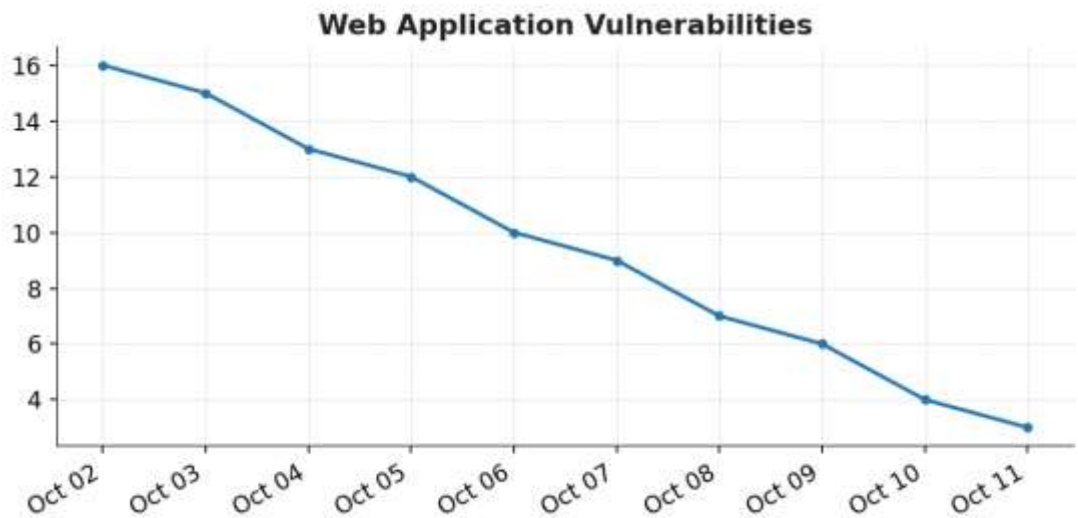
Experiment 24

Web Application Security Vulnerability Simulation

R Code:

```
x <- xts(cbind(
  Vulnerabilities = c(16,15,13,12,10,9,7,6,4,3)
), order.by=as.Date("2026-10-02")+0:9)
dygraph(x, main="Web Application Vulnerabilities")
```

Output:



Experiment 25

Security Policy Enforcement using Rule-Based Simulation

R Code:

```
x <- xts(cbind(
  Allowed = c(8,9,10,11,12,13,14,15,16,17),
  Blocked = c(7,6,6,5,4,4,3,2,2,1)
), order.by=as.Date("2026-10-05")+0:9)
dygraph(x, main="Rule-Based Security Policy")
```

Output:

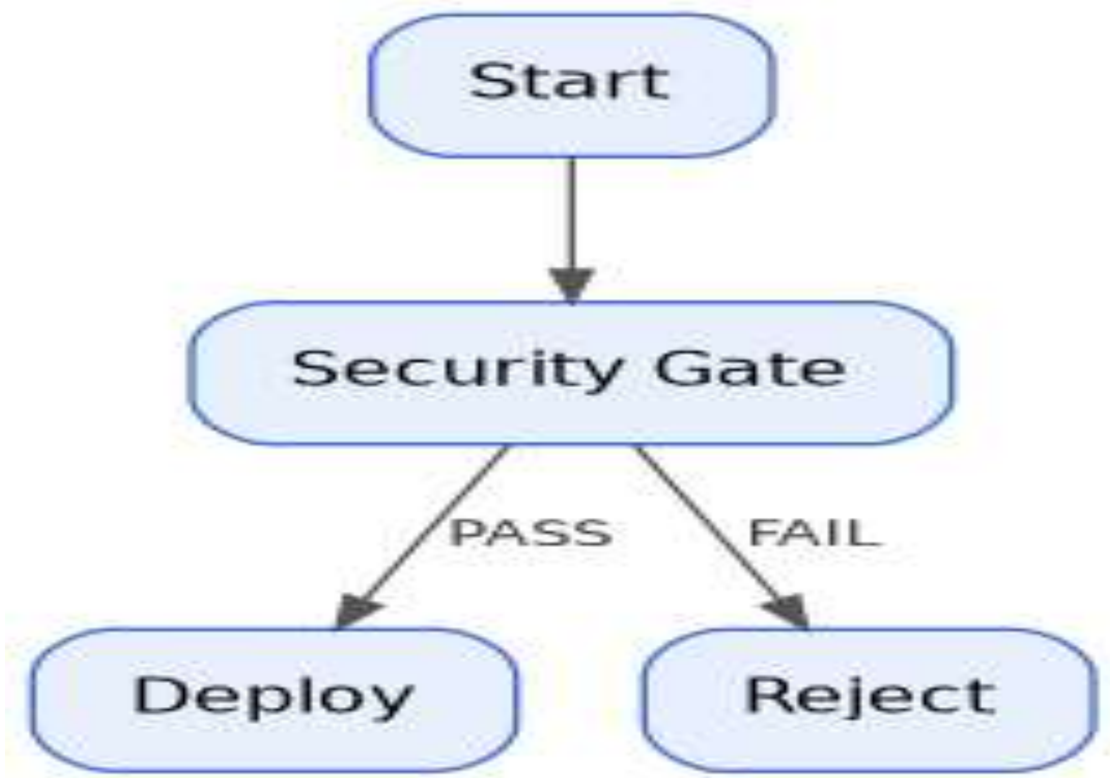
Experiment 26

DevSecOps Security Gate Modelling and Decision Simulation

R Code:

```
grViz("
digraph G {
  Start -> Security_Gate
  Security_Gate -> Deploy [label='PASS']
  Security_Gate -> Reject [label='FAIL']
}
")
```

Output:



Experiment 27

Security Metrics Generation and KPI Analysis

R Code:

```
x <- xts(cbind(
  SecurityKPI = c(60,63,66,68,71,74,76,79,82,85)
), order.by=as.Date("2026-10-09")+0:9)
dygraph(x, main="Security KPI Trend")
```

Output:

Experiment 28

Continuous Security Monitoring and Vulnerability Detection Simulation

R Code:

```
x <- xts(cbind(
  Detected = c(4,6,5,8,10,7,6,5,3,2)
), order.by=as.Date("2026-10-12")+0:9)
dygraph(x, main="Continuous Security Monitoring")
```

Output:

Experiment 29

Security Event Correlation and Attack Pattern Analysis

R Code:

```
x <- xts(cbind(
  Events = c(4,5,6,15,7,6,18,8,5,4),
  Correlated = c(1,2,2,8,3,3,10,4,2,1)
), order.by=as.Date("2026-10-14")+0:9)
dygraph(x, main="Security Event Correlation")
```

Output:

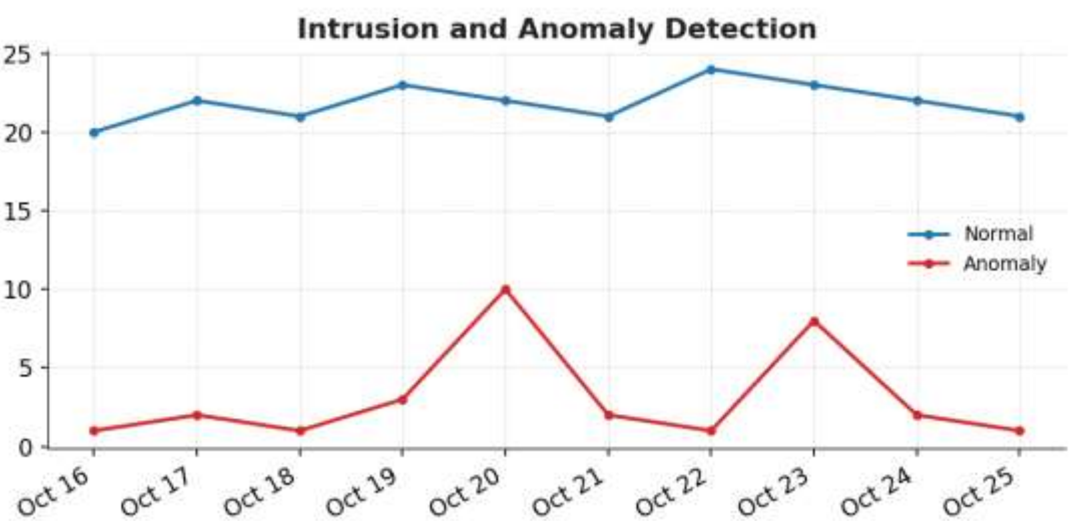
Experiment 30

Intrusion Detection and Anomaly Detection Simulation

R Code:

```
x <- xts(cbind(
  Normal = c(20,22,21,23,22,21,24,23,22,21),
  Anomaly = c(1,2,1,3,10,2,1,8,2,1)
), order.by=as.Date("2026-10-16")+0:9)
dygraph(x, main="Intrusion and Anomaly Detection")
```

Output:



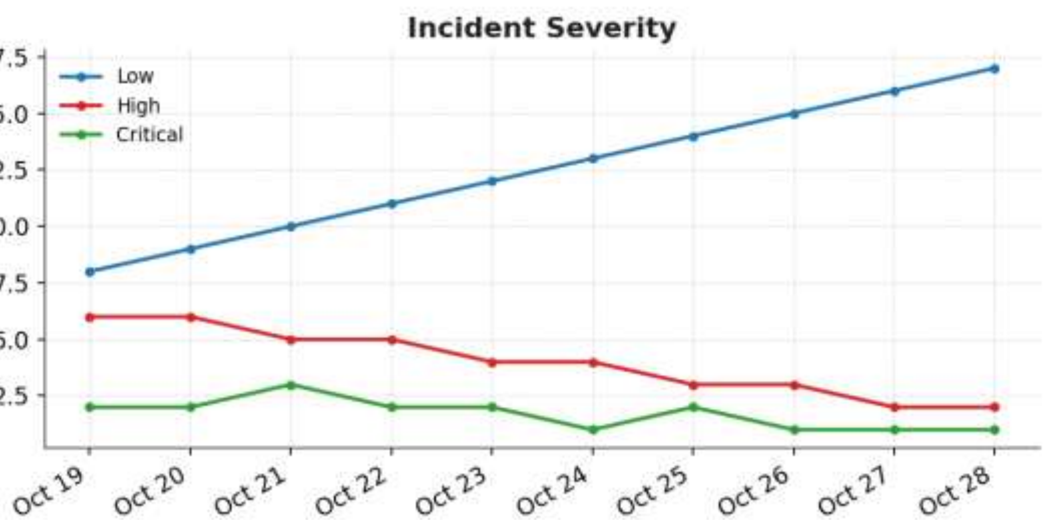
Experiment 31

Security Incident Classification and Severity Analysis

R Code:

```
x <- xts(cbind(
  Low = c(8,9,10,11,12,13,14,15,16,17),
  High = c(6,6,5,5,4,4,3,3,2,2),
  Critical = c(2,2,3,2,2,1,2,1,1,1)
), order.by=as.Date("2026-10-19")+0:9)
dygraph(x, main="Incident Severity")
```

Output:



Experiment 32

Incident Response Workflow Modelling and Simulation

R Code:

```
grViz("
digraph G {
  rankdir=LR
  Detect -> Classify -> Contain -> Eradicate -> Recover
}
")
```

Output:

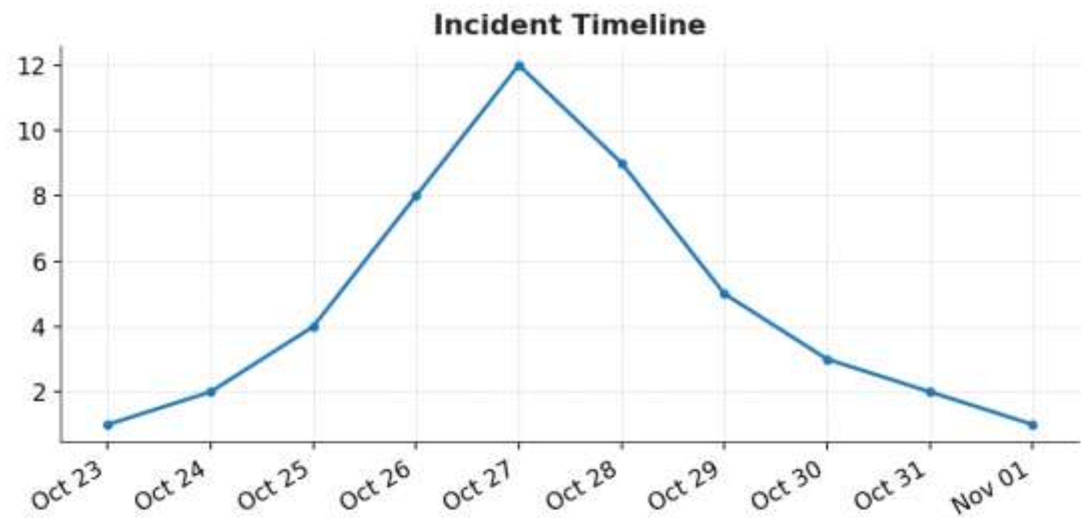
Experiment 33

Security Incident Timeline Reconstruction and Root Cause Analysis

R Code:

```
x <- xts(cbind(
  IncidentEvents = c(1,2,4,8,12,9,5,3,2,1)
), order.by=as.Date("2026-10-23")+0:9)
dygraph(x, main="Incident Timeline")
```

Output:



Experiment 34

MTTD and MTTR Analysis

R Code:

```
x <- xts(cbind(
  MTTD = c(20,18,17,15,14,12,11,10,9,8),
  MTTR = c(40,38,35,33,30,28,25,22,20,18)
), order.by=as.Date("2026-10-26")+0:9)
dygraph(x, main="MTTD and MTTR")
```

Output:

Experiment 35

Security Compliance Monitoring and Governance Simulation

R Code:

```
x <- xts(cbind(
  Compliance = c(70,72,75,77,80,82,84,87,90,92)
), order.by=as.Date("2026-10-28")+0:9)
dygraph(x, main="Security Compliance")
```

Output:

Experiment 36

DevSecOps Risk-Based Decision Making using Synthetic Data

R Code:

```
x <- xts(cbind(
  Risk = c(80,75,70,65,60,55,50,45,40,35)
), order.by=as.Date("2026-10-30")+0:9)
dygraph(x, main="Risk-Based Decision Making")
```

Output:

Experiment 37

Monte Carlo Simulation for DevSecOps Security Risk Analysis

R Code:

```
set.seed(1)
risk <- rnorm(100, mean=50, sd=10)
x <- xts(
  cbind(Risk=sort(risk)),
  order.by=as.Date("2026-11-02")+0:99
)
dygraph(x, main="Monte Carlo Risk Simulation")
```

Output:

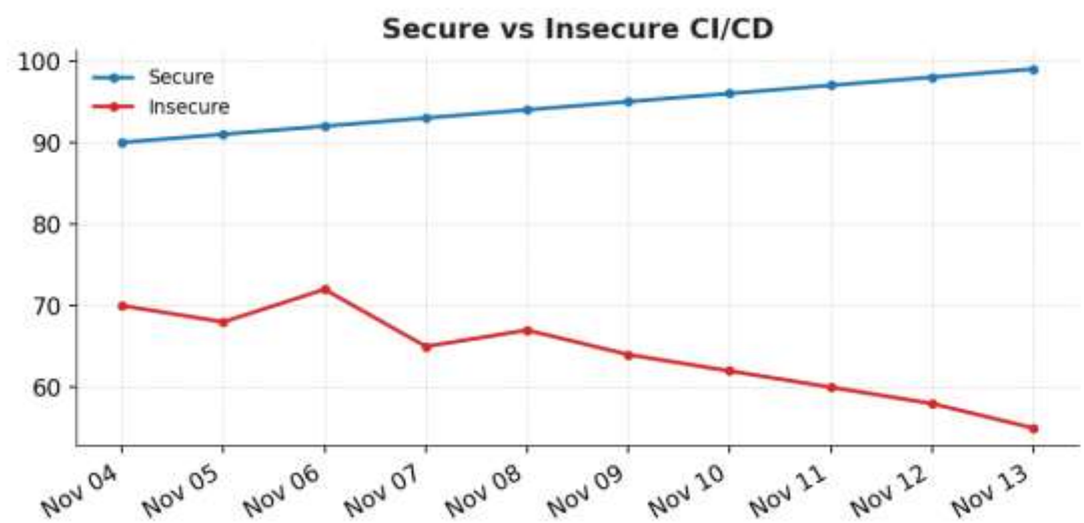
Experiment 38

Comparative Evaluation of Secure and Insecure CI/CD Pipelines

R Code:

```
x <- xts(cbind(
  Secure = c(90,91,92,93,94,95,96,97,98,99),
  Insecure = c(70,68,72,65,67,64,62,60,58,55)
), order.by=as.Date("2026-11-04")+0:9)
dygraph(x, main="Secure vs Insecure CI/CD")
```

Output:



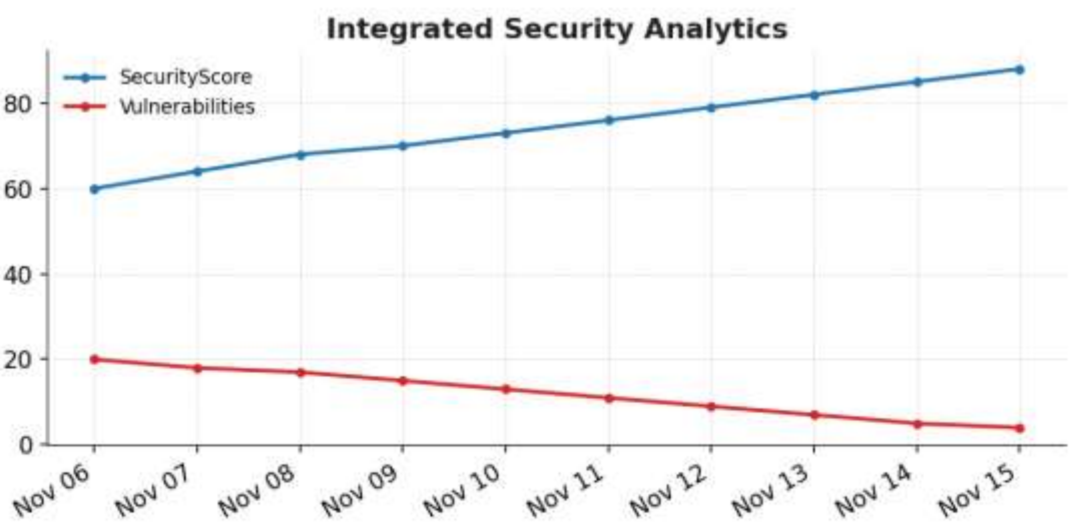
Experiment 39

Integrated DevSecOps Security Analytics and Reporting

R Code:

```
x <- xts(cbind(
  SecurityScore = c(60,64,68,70,73,76,79,82,85,88),
  Vulnerabilities = c(20,18,17,15,13,11,9,7,5,4)
), order.by=as.Date("2026-11-06")+0:9)
dygraph(x, main="Integrated Security Analytics")
```

Output:



Experiment 40

End-to-End DevSecOps Pipeline Security Simulation and Evaluation

R Code:

```
x <- xts(cbind(
  Build = c(70,72,75,78,80,82,84,86,88,90),
  Security = c(60,64,68,72,75,78,81,84,87,90),
  Deployment = c(55,60,65,68,72,76,80,84,87,91)
), order.by=as.Date("2026-11-09")+0:9)
dygraph(x, main="End-to-End DevSecOps Pipeline")
```

Output: